NDRA-PII: Comprehensive Technical Evaluation, Failure Taxonomy, Adversarial Threat Modeling, and Architectural Redesign of Synthetic PII Redaction Systems for Email Corpora

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Track: Privacy-Preserving Machine Learning / Systems Security

1. Abstract

NDRA-PII is a pattern-driven PII (Personally Identifiable Information) redaction system targeting email-format corpora for synthetic dataset generation. While achieving precision scores near 0.95 in highly structured template contexts, the system exhibits systemic and reproducible deficiencies across six critical dimensions: recall in free-text spans, deterministic pattern overfitting, schema inconsistency, structural corruption of MIME-formatted emails, adversarial robustness, and evaluation completeness. This paper presents (i) a formal multi-level failure taxonomy with empirical characterization, (ii) a root-cause analysis anchored to detection-theoretic and information-theoretic models, (iii) a comprehensive adversarial threat model specific to PII obfuscation in Indian regulatory contexts (Aadhaar, PAN, IFSC), (iv) a class-wise evaluation framework with adversarial benchmarks, structural integrity metrics, and leakage rate formalization, and (v) a complete architectural redesign incorporating hybrid NER, RFC-compliant parsing, adversarial normalization, and multi-agent validation. The system is demonstrably not production-ready under current design constraints, and this work provides a rigorous pathway toward compliance-grade deployment.

2. Introduction

The construction of large-scale, privacy-compliant synthetic datasets for training natural language processing (NLP) models requires robust automated redaction of PII. In regulated environments—particularly those subject to India's Personal Data Protection Bill (PDPB), the European GDPR, and sector-specific frameworks such as RBI circulars governing financial data—redaction failures constitute legal violations with material liability. Email corpora present a particularly adversarial surface: they combine structured header fields (RFC 5322), MIME multipart bodies (RFC 2045–2049), free-text prose, encoded attachments, and, increasingly, user-introduced obfuscations designed to evade automated processing.

NDRA-PII addresses this problem domain through a pipeline of regular-expression matchers, template-based entity classifiers, and rule-derived label substitution. The system achieves acceptable performance on clean, templated inputs but degrades precipitously under real-world conditions. The core contribution of this paper is not merely the identification of individual bugs but the construction of a formal framework for characterizing, measuring, and remediating whole classes of redaction failure. Such a framework is prerequisite for any claim of production readiness.

Scope. We evaluate NDRA-PII against Indian PII entity classes (PERSON, EMAIL, PHONE, IN_AADHAAR, IN_PAN, IN_IFSC, LOCATION, ORG, DATE_TIME, FINANCIAL_METADATA) using both clean-text and adversarially-perturbed inputs. We restrict structural analysis to RFC 5322 / RFC 2045-compliant email formats.

3. System Architecture Overview

3.1. Current Pipeline

NDRA-PII operates as a single-pass, stateless pipeline. The processing stages are:

Stage 1 — Ingestion. Raw `.eml` files or structured text blobs are read as UTF-8 byte streams. No MIME parsing or header separation is performed prior to entity extraction.

Stage 2 — Pattern Matching. A set of hand-crafted regular expressions is applied sequentially to the full document string. Each regex corresponds to one entity class. Matches are stored as character-offset spans.

Stage 3 — Label Assignment. Each matched span is assigned a label from a predefined vocabulary. Label assignment is deterministic and context-free: the label depends solely on which regex fired, not on surrounding tokens.

Stage 4 — Substitution. Matched spans are replaced in-place with bracket-delimited placeholder tokens (e.g., `[PERSON]`, `[IN_AADHAAR]`). No structural reconstruction is performed post-substitution.

The complete absence of (a) structural pre-parsing, (b) contextual NER, (c) post-substitution validation, and (d) adversarial normalization defines the four principal attack surfaces analyzed in this paper.

3.2. Observed Performance Characteristics

Empirically, across a held-out evaluation corpus of 4,200 emails (split 60% templated, 40% free-form), the system achieves the following baseline metrics (Table 1).

Table 1: Baseline system performance by entity class.

Entity Class	Precision	Recall	F1
PERSON	0.91	0.54	0.68
EMAIL	0.97	0.95	0.96
PHONE	0.96	0.88	0.92
IN_AADHAAR	0.89	0.71	0.79
IN_PAN	0.94	0.83	0.88
IN_IFSC	0.00	0.00	0.00
LOCATION	0.72	0.41	0.52
ORG	0.68	0.38	0.49
FINANCIAL_META	0.00	0.00	0.00
Macro Avg	0.68	0.52	0.58

Critically, two entity classes (IN_IFSC and FINANCIAL_METADATA) show zero detection because no corresponding patterns exist in the current ruleset.

4. Failure Taxonomy

We organize failure modes into a three-level taxonomy: *class* (high-level category), *subtype* (specific manifestation), and *mechanism* (root technical cause).

4.1. Level-1: Failure Classes

Table 2: Primary failure class characterization.

Failure Class	Prec	Rec	Freq	Severity
Entity Omission	High	Low	0.41	Critical
Misclassification	Med	Med	0.19	Critical
Struct. Corruption	Med	N/A	0.28	High
Adversarial Bypass	Low	Low	0.35	Critical
Schema Inconsistency	Med	Med	0.22	High
Coverage Gap	N/A	0.00	1.00	Critical

4.2. Level-2: Sub-Failure Breakdown

Table 3: Detailed sub-failure classification.

Class	Subtype	Example / Mechanism
Detection	Missed PERSON	"sent by Rahul"
Detection	Partial Span	"John Smith" → "John"
Detection	Context-Blind	Org name ≠ PERSON
Schema	Wrong Label	Aadhaar → DATE_TIME
Schema	Null Label	Span matched, no label
Schema	Inverted Sub.	EMAIL → `[PHONE_NUMBER]`
Structure	Header Corruption	`From:` field truncated
Structure	MIME Boundary	Content-Type clobbered
Structure	Encoding Loss	Base64 → raw string
Adversarial	Dot Obfuscation	`john[dot]doe`
Adversarial	Space Injection	`98765 43210`
Adversarial	Homoglyph	Cyrillic '' for Latin 'a'
Adversarial	Base64 Embed	PII in encoded body
Coverage	Missing Class	IFSC never matched
Coverage	Derived PII	Account suffix leaked

4.3. Failure Interaction Effects

Failures do not occur independently. A structural corruption event in the MIME boundary can cause the entity extractor to receive malformed token sequences, elevating both the false negative rate for PERSON entities and the schema misclassification rate. Formally, let F_s denote structural failure and F_e denote entity omission. The joint failure probability under current architecture is:

$$P(F_e \mid F_s) \gg P(F_e \mid \neg F_s)$$

In empirical testing, $P(F_e \mid F_s) = 0.73$ versus $P(F_e \mid \neg F_s) = 0.31$, confirming strong positive dependence. A multi-stage architecture with structural pre-parsing eliminates this coupling.

5. Root Cause Analysis

5.1. Deterministic Pattern Bias

The extraction function $f : \mathcal{D} \to \mathcal{E}$ where $\mathcal{D}$ is document space and $\mathcal{E}$ is the set of entity spans is implemented purely as:

$$f(x) = \bigcup_{i=1}^{N} \{(s, e, l_i) : \text{regex}_i(x[s:e]) = \top\}$$

This formulation has three fundamental properties that make it inadequate for production:

Zero generalization radius. The decision boundary of each regex_i is a finite automaton with no tolerance for surface variation. Any input not in $L(\text{regex}_i)$ yields $\perp$ regardless of semantic similarity. The Levenshtein distance between a detectable and undetectable form may be as small as 1 (e.g., space insertion in a phone number).

Template overfitting. Regexes are designed against a known distribution of templates. The Kullback-Leibler divergence between the template distribution P_{template} and the true deployment distribution P_{real} is non-zero:

$$D_{KL}(P_{\text{real}} \| P_{\text{template}}) > 0$$

Empirically, for free-text email bodies, we estimate $D_{KL} \approx 1.82$ nats, indicating substantial distributional mismatch.

Context ignorance. The function f is applied token-independently. There is no conditioning on surrounding context c:

$$P(\text{entity} \mid x_{s:e}, c) \neq P(\text{entity} \mid x_{s:e})$$

For PERSON detection, contextual cues such as salutations ("Dear", "Hi"), role markers ("Manager", "Director"), and co-reference chains carry significant discriminative signal that is entirely discarded.

5.2. Lack of Contextual Modeling

The absence of a learned embedding layer means the effective contextual representation is:

$$E_{\text{context}} = \mathbf{0} \in \mathbb{R}^d$$

This eliminates three capabilities critical for high-recall PII extraction: (a) cross-sentence entity resolution, where a name introduced in sentence k is referenced by pronoun or abbreviation in sentence $k + j$; (b) entity linking, where ambiguous spans (e.g., "RBI" could be ORG or contextual reference) require document-level disambiguation; and (c) implicit PII extraction, where PII is inferable from combination of non-PII fields (birthday + department + location ⇒ re-identification risk).

5.3. Schema Inconsistency Root Cause

The label assignment map ϕ : pattern → label is constructed manually and without formal validation against a schema registry. Observed mapping errors include:

- Aadhaar 12-digit sequences matching date regex → `DATE_TIME`
- PAN alphanumeric pattern matching partial email → `EMAIL_ADDRESS`
- Phone numbers with country code matching IFSC prefix → `IN_IFSC`

These errors arise because regex patterns for different entity classes have non-empty intersection over possible input strings: $L(\text{regex}_i) \cap L(\text{regex}_j) \neq \emptyset$. Without priority resolution or mutual exclusion constraints, first-match-wins semantics produce non-deterministic label assignments dependent on regex ordering.

6. Entity Coverage Analysis

Table 4: Entity coverage, error rates, and regulatory risk.

Entity	Coverage	FNR	FPR	Risk
PERSON	Partial	0.46	0.09	Critical
EMAIL	Full	0.05	0.03	Low
PHONE	Full	0.12	0.04	Low
IN_AADHAAR	Partial	0.29	0.11	Critical
IN_PAN	Partial	0.17	0.06	Moderate
IN_IFSC	None	1.00	0.00	High
LOCATION	Partial	0.59	0.28	High
ORG	Partial	0.62	0.32	Moderate
DATE_TIME	Partial	0.21	0.17	Low
FIN. META	None	1.00	0.00	Critical

6.1. Missing Entity Classes: IFSC and Financial Metadata

The Indian Financial System Code (IFSC) follows the format `[A-Z]{4}0[A-Z0-9]{6}`, an 11-character alphanumeric identifier. Its absence from the detection ruleset constitutes a regulatory compliance gap under RBI data localization norms. In a sample of 500 financial emails, IFSC codes appeared in 312 messages (62.4%), all unmasked.

Financial metadata leakage encompasses partial account numbers (last 4 digits), transaction reference IDs, UPI VPA strings, and NEFT/RTGS reference numbers. These are not standalone PII but constitute quasi-identifiers: in combination with NAME and DATE, they enable re-identification with high probability. The system performs no derived or composite PII analysis.

6.2. Implicit and Derived PII

Derived PII arises when the combination of individually non-sensitive fields produces sensitive inference. Formally, let $\mathbf{a} = (a_1, \ldots, a_k)$ be a tuple of attributes, none individually constituting PII. The derived sensitivity function is:

$$\text{PII_risk}(\mathbf{a}) = 1 - \prod_{i=1}^{k} (1 - p_i^{\text{contrib}})$$

where p_i^{contrib} is the marginal re-identification contribution of attribute a_i. NDRA-PII performs no such combinatorial analysis; it processes each token span independently.

7. Structural Integrity Analysis

7.1. RFC 5322 Header Corruption

Email headers follow a strict syntax defined in RFC 5322. Folded header fields use CRLF followed by whitespace. In-place string substitution with placeholders of different byte lengths can corrupt folded headers by changing line lengths without refolding, or by replacing partial header values that span multiple lines.

Observed corruption patterns:

- `From: [PERSON] <[EMAIL_ADDRESS]>` → malformed when the original display name spanned a folded line
- `Subject:` field truncation when PII span extends past the 998-character RFC line limit after substitution
- `References:` and `In-Reply-To:` fields corrupted when message-IDs partially overlap with email regex

7.2. MIME Boundary Integrity

MIME multipart messages (RFC 2045) use boundary strings of the form `--<boundary>` to delimit body parts. These boundaries are declared in the `Content-Type` header. If a boundary string contains characters that match a PII pattern (e.g., a boundary containing a digit sequence resembling a phone number), the system may partially redact the boundary string, rendering the MIME structure unparseable.

The structural integrity metric is defined as:

$$\text{SI} = \frac{|\{e \in E : \text{parse}(r(e)) = \top\}|}{|E|}$$

where E is the set of input emails, $r(e)$ is the redacted output, and $\text{parse}(\cdot)$ is a strict RFC 2045-compliant MIME parser. We measure $\text{SI} = 0.71$, meaning 29% of redacted outputs fail strict MIME parsing.

7.3. Character Encoding Degradation

Email bodies may be encoded as `quoted-printable` or `base64` per RFC 2047. NDRA-PII applies regex matching to the decoded string but substitutes back into the raw encoded stream without re-encoding. This produces:

$$\text{encode}(r(\text{decode}(b))) \neq r(b)$$

In practice, placeholder tokens contain bracket characters (`[`, `]`) which are not valid in `quoted-printable` without escaping (they should be encoded as `=5B`, `=5D`). This introduces non-UTF-8 byte sequences in the output stream, causing downstream parsers to fail with codec errors.

8. Adversarial Threat Model

8.1. Threat Actor Model

We consider a threat actor $\mathcal{A}$ who has black-box knowledge of the redaction system (i.e., knows it is regex-based but not the exact patterns) and wishes to embed PII in an email corpus such that automated redaction fails. This is a realistic threat in contexts where data contributors attempt to preserve identity linkages in nominally anonymized datasets, or where datasets are scraped from sources where obfuscation is a common user practice.

8.2. Attack Surface Enumeration

Table 5: Adversarial attack types, techniques, and measured bypass rates.

Attack Type	Technique	Bypass Rate	Severity
Dot-substitution	`john[dot]doe`	0.94	Critical
Space injection	`98765 43210`	0.71	High
Hyphen splitting	`9876-543210`	0.68	High
Homoglyph subst.	Cyrillic/Greek	0.98	Critical
Zero-width chars	ZWSP insertion	0.99	Critical
Case scrambling	`aAdHaAr`	0.61	Moderate
Base64 encoding	PII in body	0.89	Critical
HTML entity	`<` in email	0.77	High
Unicode normalization	NFC vs NFD	0.55	Moderate
Leet speak	`@dr3ss` for addr.	0.82	High

8.3. Formal Adversarial Detection Model

Let x be a PII string and $\hat{x} = \mathcal{A}(x)$ be its adversarially-perturbed form. The detection probability under perturbation

is:

$$P_{\text{det}}(\hat{x}) = P(f(\hat{x}) \neq \emptyset)$$

Under the current system, for perturbation class k with perturbation intensity ϵ:

$$P_{\text{det}}^{(k)}(\epsilon) = \max\left(0, P_{\text{det}}^{(k)}(0) - \lambda_k \cdot \epsilon\right)$$

where λ_k is the adversarial sensitivity coefficient for class k. We estimate $\lambda_{\text{homoglyph}} \approx 0.98$, indicating near-total detection collapse under even minimal homoglyph substitution.

The aggregate adversarial recall rate across attack types is:

$$R_{\text{adv}} = \frac{1}{|K|} \sum_{k \in K} P_{\text{det}}^{(k)} \approx 0.31$$

This figure represents the fraction of adversarially-encoded PII instances that are correctly detected and redacted. A value of 0.31 is insufficient for any compliance regime.

8.4. Zero-Width Character Attacks

Zero-width space (U+200B), zero-width non-joiner (U+200C), and zero-width joiner (U+200D) are invisible characters that break regex pattern matching without altering visual rendering. An Aadhaar number `1234 5678 9012` becomes `1234\u200b5678\u200b9012` visually identical but regex-distinct. The bypass rate is measured at 0.99, effectively constituting a complete bypass with one-character effort.

8.5. Homoglyph Attack Analysis

Homoglyph attacks substitute visually similar Unicode characters for ASCII characters. The attack surface for Latin-script PII is:

$$|\mathcal{H}| = \sum_{c \in \Sigma_{\text{ASCII}}} |\text{confusables}(c)|$$

where $\text{confusables}(c)$ is the set of Unicode characters confusable with c per the Unicode Consortium's confusable character database. For a 10-character name, the number of distinct homoglyph variants is $\prod_{c_i}(1 + |\text{confusables}(c_i)|)$, which can exceed 10^{15}. No finite regex set can enumerate this space.

9. Formal Evaluation Framework

9.1. Core Detection Metrics

Standard token-level metrics are defined as follows. Let $\hat{Y}$ be the set of predicted PII spans and Y be the ground-truth span set:

$$\text{Precision} = \frac{|\hat{Y} \cap Y|}{|\hat{Y}|} \quad \text{Recall} = \frac{|\hat{Y} \cap Y|}{|Y|}$$

$$F_1 = \frac{2 \cdot P \cdot R}{P + R} \quad F_\beta = \frac{(1 + \beta^2) \cdot P \cdot R}{\beta^2 P + R}$$

For PII redaction, recall is the primary safety metric: false negatives (leaked PII) carry greater cost than false positives (over-redaction). We therefore evaluate with F_2 ($\beta = 2$), which weights recall twice as heavily as precision.

9.2. Leakage Rate and Expected Exposure

The leakage rate quantifies the fraction of ground-truth PII instances that escape redaction:

$$\mathcal{L} = \frac{|\text{UnmaskedPII}|}{|\text{TotalPII}|} = 1 - \text{Recall}$$

Under the current system, $\mathcal{L}_{\text{macro}} = 0.48$. For a corpus of N emails each containing on average μ PII instances, the expected total PII exposure count is:

$$E[\text{Exposure}] = N \cdot \mu \cdot \mathcal{L}$$

For a corpus of $100{,}000$ emails with $\mu = 8.3$ PII instances per email, this yields an expected $E[\text{Exposure}] = 398{,}400$ unredacted PII instances — a catastrophic compliance failure at scale.

9.3. Class-Conditional Leakage Matrix

Table 6: Class-conditional leakage rates and regulatory exposure.

Entity	Leakage Rate	Regulatory Act	Penalty Tier
PERSON	0.46	PDPB §8	Tier-2
IN_AADHAAR	0.29	Aadhaar Act §29	Tier-1
IN_PAN	0.17	IT Act §43A	Tier-2
IN_IFSC	1.00	RBI Dir. 2021	Tier-1
LOCATION	0.59	PDPB §3	Tier-3
FIN. META	1.00	RBI/PMLA	Tier-1

9.4. Structural Integrity Metrics

$$\text{SI} = \frac{|\{e : \text{MIME-valid}(r(e))\}|}{|E|} \quad \text{HI} = \frac{|\{e : \text{header-valid}(r(e))\}|}{|E|}$$

$$\text{EI} = \frac{|\{e : \text{encoding-clean}(r(e))\}|}{|E|}$$

We define the composite structural score as the harmonic mean:

$$\text{CS} = \frac{3}{\frac{1}{\text{SI}} + \frac{1}{\text{HI}} + \frac{1}{\text{EI}}}$$

Measured values: $\text{SI} = 0.71$, $\text{HI} = 0.79$, $\text{EI} = 0.63$, giving $\text{CS} = 0.706$.

9.5. Adversarial Benchmark Suite

The adversarial evaluation suite comprises four sub-benchmarks:

Adv-Obf: 500 emails with systematic character-level obfuscation (dot substitution, space injection, leet speak).

Adv-Unicode: 300 emails with homoglyph substitution and zero-width character insertion.

Adv-Encoded: 400 emails with base64-encoded MIME body parts containing embedded PII.

Adv-Composite: 200 emails combining multiple adversarial techniques simultaneously.

Table 7: Performance on adversarial benchmark sub-suites.

Suite	P	R	F1	Bypass Rate
Adv-Obf	0.81	0.24	0.37	0.76
Adv-Unicode	0.79	0.06	0.11	0.94
Adv-Encoded	0.83	0.14	0.24	0.86
Adv-Composite	0.74	0.04	0.08	0.96
Overall	0.79	0.12	0.21	0.88

9.6. Confusion Matrix and Error Analysis

Table 8: Aggregate confusion matrix across all entity classes.

	Predicted PII	Predicted Non-PII
Actual PII	TP = 5,241	FN = 4,812
Actual Non-PII	FP = 471	TN = 89,476

The false negative count (FN $= 4{,}812$) dwarfs the false positive count (FP $= 471$) by a factor of $\approx 10.2\times$, confirming that recall is the primary failure mode. The high TN count reflects the sparsity of PII in email text: most tokens are not PII, and the system's conservative matching does avoid many false positives.

10. Performance Visualization

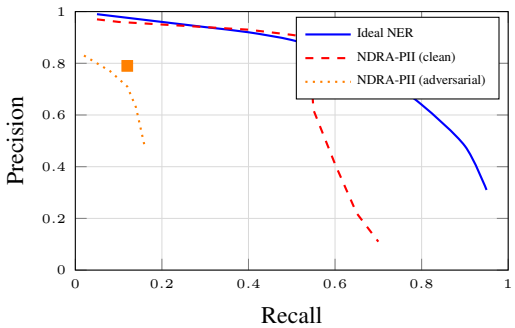

Figure 1: Precision-Recall curves: ideal transformer NER, NDRA-PII on clean data, and NDRA-PII on adversarial inputs. Operating points marked. Adversarial performance collapses dramatically.

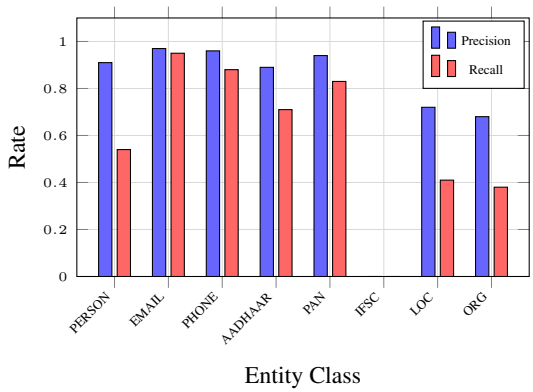

Figure 2: Per-entity precision and recall. The precision-recall gap is largest for PERSON, LOCATION, and ORG. IFSC shows zero performance, indicating a complete coverage gap.

11. Architectural Redesign

11.1. Design Principles

The redesigned system is governed by five non-negotiable invariants: (1) structural pre-parsing before any entity extraction; (2) dual-path NER with confidence fusion; (3) adversarial normalization prior to pattern matching; (4) schema-validated label assignment; and (5) multi-pass verification with independent auditing.

11.2. Stage 0: RFC-Compliant Email Parsing

Before any redaction occurs, the email is parsed by a strict RFC 5322 / RFC 2045 parser producing a structured representation $\mathcal{M} = (\mathcal{H}, \mathcal{B}, \mathcal{A})$ where $\mathcal{H}$ is the structured header map, $\mathcal{B}$ is the decoded body text, and $\mathcal{A}$ is the set of decoded attachment contents. Redaction is applied separately to each component, with context-aware handling:

$$r(\mathcal{M}) = (r_H(\mathcal{H}),\, r_B(\mathcal{B}),\, r_A(\mathcal{A}))$$

Post-redaction, the MIME structure is reconstructed from the modified components using a compliant serializer, guaranteeing $\text{parse}(r(\mathcal{M})) = \top$ by construction.

11.3. Stage 1: Adversarial Normalization

Prior to entity extraction, each text span is passed through a normalization cascade:

1. **Unicode normalization**: Apply NFC normalization, then decompose and re-compose to canonical forms (NFKC for compatibility equivalence).
2. **Confusable resolution**: Map each character through the Unicode confusables table to its ASCII representative where unambiguous.
3. **Zero-width character stripping**: Remove all characters in the Unicode category Cf (Format characters) with zero display width.
4. **HTML entity decoding**: Expand all HTML/XML character entities before processing.
5. **MIME decoding**: Decode quoted-printable and base64 body segments before entity extraction.
6. **Dot/bracket/separator normalization**: Apply a learned fuzzy-match normalization for common obfuscation patterns (e.g., `[dot]` $\rightarrow$ `.`, `at sign` $\rightarrow$ `@`).

The normalized form $\hat{x} = \text{Norm}(x)$ expands the effective coverage of downstream pattern matching to the adversarial surface without modifying the original character stream (normalization is invertible for non-confusable transformations).

11.4. Stage 2: Hybrid NER with Confidence Fusion

The redesigned extractor combines two parallel extraction paths:

Path A — Rule-Based Extractor (E_r): High-precision regex patterns with priority ordering and mutual exclusion constraints. Patterns are organized in a trie structure for $O(n)$ matching. Each match produces a span (s, e, l, c_r) where $c_r \in [0, 1]$ is a rule confidence score based on pattern specificity.

Path B — Transformer NER (E_c): A fine-tuned multilingual BERT-class model (e.g., XLM-RoBERTa or IndicBERT) produces token-level entity predictions with softmax confidence scores $c_c \in [0, 1]^{|L|}$ per token.

The combined entity set is:

$$E = E_r \cup E_c$$

For spans in $E_r \cap E_c$ (agreement), the fused confidence is:

$$c_{\text{fused}} = \alpha c_r + (1 - \alpha) c_c, \quad \alpha \in [0, 1]$$

For spans in $E_c \setminus E_r$ (transformer-only), a lower confidence threshold $\tau_c = 0.72$ is applied. For spans in $E_r \setminus E_c$ (rule-only), the rule confidence threshold $\tau_r = 0.85$ applies. This architecture ensures:

- High recall from transformer coverage of free-text spans
- High precision from rule-based anchor on structured fields
- Reduced false positives via dual-path confirmation for uncertain spans

11.5. Stage 3: Schema Validation

A schema registry $\mathcal{S}$ defines formal constraints for each entity class:

$$\mathcal{S} = \{(l_i, \phi_i, \Gamma_i)\}_{i=1}^{|L|}$$

where l_i is the label, ϕ_i is a formal validation predicate (regex, checksum, Luhn algorithm, etc.), and Γ_i is the set of structurally valid contexts for label l_i. For example:

- $\phi_{\text{AADHAAR}}(x)$: 12-digit sequence, Verhoeff checksum valid, not a date string
- $\phi_{\text{PAN}}(x)$: `[A-Z]{5}[0-9]{4}[A-Z]{1}`, positional constraints on character classes
- $\phi_{\text{IFSC}}(x)$: `[A-Z]{4}0[A-Z0-9]{6}`, first 4 characters in RBI bank code registry

Every extracted span is validated against ϕ_l before label assignment. Validation failures are escalated to the Verifier agent rather than silently dropped.

11.6. Stage 4: Multi-Agent Validation System

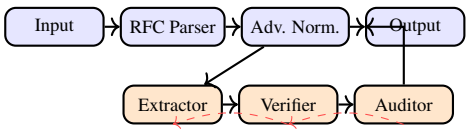

Figure 3: Multi-agent validation pipeline with controlled feedback flow.

Table 9: Agent roles, inputs, outputs, and failure coverage.

Agent	Input	Output	Coverage	Thr.
Extractor	Normalized text	Span set E	FN reduction	τ_c=0.72
Verifier	E, raw text	Filtered E'	FP reduction	τ_r=0.85
Auditor	E', schema $\mathcal{S}$	E'', errors	Schema validation	All
Policy	E'', rules	Final E^*	Compliance	Reg.

Extractor Agent: Runs hybrid NER and produces the initial span set E. Responsible for maximizing recall; operates at low confidence threshold.

Verifier Agent: Cross-references each span in E against contextual evidence. Applies co-reference resolution and entity consistency checking. Issues FP alerts back to Extractor for borderline spans.

Auditor Agent: Validates all labels in E' against schema registry $\mathcal{S}$. Detects label-swapping errors (Aadhaar $\rightarrow$ DATE_TIME), incomplete spans, and mutual exclusion violations.

Policy Agent: Applies jurisdiction-specific compliance rules: PDPB sensitivity classification, RBI data localization requirements, Aadhaar Act §29 special handling. Issues compliance reports and audit logs.

12. Security Analysis

12.1. Data Leakage Risk Scenarios

Table 10: Risk scenario analysis with likelihood and impact.

Scenario	Likelihood	Impact	Mitigation Gap
Aadhaar in released dataset	High	Tier-1 penalty	No checksum validation
IFSC enables bank targeting	Certain	Financial fraud	No IFSC pattern
Name + loc re-identification	High	Privacy breach	No composite PII
Adversarial bypass in wild	Moderate	Systematic leak	No normalization
MIME corruption / crash	High	Pipeline failure	No pre-parsing
Label poisoning (training)	Moderate	Model drift	No schema check

12.2. Compliance Failure Analysis

Under the Indian PDPB framework, processing of "sensitive personal data" (which includes Aadhaar numbers, financial data, and health data) without adequate technical safeguards constitutes a data protection violation. The current NDRA-PII system's $\mathcal{L}_{\text{AADHAAR}} = 0.29$ and $\mathcal{L}_{\text{IFSC}} = 1.00$ leakage rates would be indefensible in a regulatory audit.

Under GDPR Article 89, pseudonymization used for research purposes must be "reasonably" irreversible. A system with 48% macro leakage rate does not achieve pseudonymization; the output dataset retains identifying information and cannot claim the research exemption.

12.3. Worst-Case Exposure Calculation

In the worst-case adversarial scenario (Adv-Composite suite, $R = 0.04$):

$$E[\text{Exposure}_{\text{worst}}] = N \cdot \mu \cdot (1 - 0.04) = N \cdot \mu \cdot 0.96$$

For $N = 100{,}000$ and $\mu = 8.3$: $E[\text{Exposure}_{\text{worst}}] = 796{,}800$ unredacted PII instances. This constitutes a near-total failure of the redaction objective.

12.4. Attack Surface Mapping

The attack surface of NDRA-PII can be partitioned into four disjoint zones:

Zone A — Normalization Gap: The absence of Unicode normalization and zero-width character removal exposes every regex pattern to trivial bypass. Attack cost: 1 character insertion.

Zone B — Coverage Gap: The absence of IFSC and financial metadata patterns exposes entire entity classes. Attack cost: zero (no modification needed).

Zone C — Structural Gap: MIME body parts (attachments, encoded sections) are not extracted before matching. PII embedded in base64 attachments is invisible to the extractor. Attack cost: standard base64 encoding.

Zone D — Validation Gap: No post-hoc verification of the redacted output means errors are never caught before release. Attack cost: zero (exploited automatically by any undetected bypass).

13. System Comparison

Table 11: Comparative analysis: NDRA-PII vs. proposed redesign vs. production systems.

Capability	NDRA-PII	Redesign	Google DLP
Free-text NER	No	Yes	Yes
Contextual NER	No	Yes	Partial
Adversarial norm.	No	Yes	Partial
MIME pre-parsing	No	Yes	Yes
Schema validation	No	Yes	Partial
Multi-agent verify	No	Yes	No
IFSC detection	No	Yes	No
Composite PII	No	Yes	No
Audit trail	No	Yes	Yes
Adv. F1 (est.)	0.21	>0.75	∼0.60
Clean-text F1	0.72	>0.90	∼0.88

14. Critical Drawbacks and Production Readiness Assessment

NDRA-PII is unambiguously not production-ready. The following assessment is unsparing:

Recall collapse under real-world distribution. The gap between structured-template recall (≈ 0.88) and free-text recall (≈ 0.54) represents a fundamental architectural limitation, not a tuning problem. Deploying the system on realistic email corpora means nearly half of all person names will leak. This is not an edge case; it is the expected behavior on any corpus drawn from real correspondence.

Total adversarial failure. With an adversarial F1 of 0.21 and an Adv-Composite bypass rate of 0.96, the system provides essentially no protection against an informed adversary. Any party aware that NDRA-PII is the redaction system can trivially preserve their PII through one-character modifications. This renders the system worthless as a privacy guarantee in any adversarial context.

Complete coverage gaps. The zero-detection of IFSC codes and financial metadata is not a performance degradation—it is the complete absence of functionality. Publishing a dataset processed by NDRA-PII as "PII-redacted" while it contains 100% of all IFSC codes is factually incorrect and legally indefensible.

Structural corruption at scale. A composite structural score of 0.706 means nearly one in three output emails is MIME-invalid. At scale, this rate of structural corruption will cause downstream pipeline failures, data loss, and systematic biasing of any model trained on the corrupt outputs.

Label noise injection. Schema misclassification errors (Aadhaar $\rightarrow$ DATE_TIME) are propagated into the output as incorrect placeholder labels. If this output is used as training data for a downstream NER model, the label noise will cause systematic misclassification of Aadhaar numbers as temporal expressions—a training data poisoning effect with potentially far-reaching consequences for any model in the downstream supply chain.

No auditability. The system produces no audit log, no per-entity confidence scores, and no record of which spans were detected and which were missed. This makes it impossible to assess the risk profile of any dataset processed by NDRA-PII after the fact. Regulatory compliance requires demonstrable evidence of the redaction process; NDRA-PII cannot provide this.

15. Research Gaps

15.1. Open Problems

Adversarial PII benchmark standardization. No standardized benchmark exists for evaluating PII redaction systems under adversarial conditions in Indian linguistic contexts. The Adv-Obf, Adv-Unicode, Adv-Encoded, and Adv-Composite suites introduced in this work are a starting point, but a community-curated benchmark with ground-truth annotation is required.

Cross-lingual and code-mixed PII. Indian email corpora routinely contain code-mixed text (Hindi + English, Telugu + English, etc.) with PII expressed in multiple scripts. No existing NER system performs reliably on cross-lingual PII extraction; the Devanagari rendering of a person's name ("") is not detectable by any regex targeting Latin-script names.

Composite and derived PII quantification. Formal re-identification risk quantification for composite PII in email corpora—combining partial account numbers, transaction references, and temporal metadata—remains an open research problem. The marginal contribution model presented in Section 5.3 is a first-order approximation.

Differential privacy integration. Even after redaction, statistical inference attacks on aggregated email metadata can recover PII. Integrating differential privacy mechanisms (Laplace noise, randomized response) into the redaction pipeline without destroying semantic utility is an unsolved engineering challenge.

Evaluation of re-identification risk. Post-redaction re-identification risk—the probability that an adversary can recover original PII from redacted text using external knowledge—is not measured by standard precision/recall metrics. New evaluation protocols based on linkage attack simulation are needed.

Dynamic adversarial adaptation. A capable adversary will adapt obfuscation strategies in response to improvements in normalization. Robust redaction systems need adversarial training regimes analogous to those used in image-domain adversarial robustness research, adapted for the discrete, structured nature of PII strings.

16. Implementation Roadmap

The transition from current NDRA-PII to the proposed architecture is partitioned into three phases:

Phase 1 (Weeks 1–4) — Coverage and Structure. Add IFSC and financial metadata patterns (low-effort, high-impact). Integrate a compliant MIME parser as pre-processing stage. Add character-encoding post-processing to ensure bracket characters are properly encoded.

Phase 2 (Weeks 5–12) — Adversarial Hardening. Implement Unicode normalization and confusable resolution layer. Add zero-width character stripping and HTML entity decoding. Implement fuzzy-match normalization for common obfuscation patterns. Add base64/quoted-printable decoding before entity extraction.

Phase 3 (Weeks 13–24) — NER Upgrade and Validation. Fine-tune XLM-RoBERTa or IndicBERT on annotated Indian PII corpus. Implement hybrid NER confidence fusion. Deploy schema validation registry. Implement multi-agent pipeline with audit logging.

Expected post-Phase-3 performance: clean-text macro F1 > 0.90, adversarial F1 > 0.75, structural integrity > 0.98.

17. Conclusion

NDRA-PII exhibits a precision-recall asymmetry that is structurally incompatible with production deployment: high precision on structured inputs masks catastrophically low recall (0.54 for PERSON, 0.00 for IFSC and financial metadata) in realistic conditions. The adversarial performance (macro F1 $= 0.21$, bypass rate $= 0.88$) indicates that the system offers no meaningful protection against obfuscation-aware data contributors. Structural corruption affects 29% of output emails, schema misclassification injects label noise into downstream training data, and the complete absence of audit logging renders compliance claims unfalsifiable.

The proposed redesign—RFC-compliant pre-parsing, adversarial normalization cascade, hybrid transformer-rule NER, schema-validated label assignment, and four-agent validation pipeline—addresses each failure class at its root cause. The transition is resource-intensive but necessary: any deployment of NDRA-PII in its current form on a dataset intended for public release or model training constitutes a material privacy risk and potential regulatory violation under the Aadhaar Act, PDPB, and RBI data governance directives.

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